

RESUSCITATIVE THORACOTOMY

SFM CSOG 010 'RESUSCITATIVE THORACOTOMY' 2022.

ONLY PERFORM IF SURGICAL FACILITY IS AVAILABLE
<1 HR FOR PRIMARY AND <30 MIN FOR SECONDARY

PRIMARY INDICATION

PENETRATING TRAUMA TO TORSO OR NECK

SECONDARY INDICATION

SUBDIAPHRAGMATIC HAEMORRHAGE WITHOUT TBI

& LOSS OF VITAL SIGNS (CENTRAL PULSE, MENTATION & RESP EFFORT) <15 MINUTES
& HAS ETCO₂ ABOVE >1.0kPa OR ECG ACTIVITY OR CARDIAC MOVEMENT ON ULTRASOUND

360 ACCESS, RAPID HEAD TO TOE and EXCLUDE FUTILITY (ROLE CRITERIA page 7)

PERFORM BILATERAL THORACOSTOMIES & GET KIT DUMP READY

ANY IMPROVEMENT TO CIRCULATION? (INCREASED ETCO₂, CENTRAL PULSES PRESENT)

NO

CONTINUE WITH THORACOTOMY

YES

CONTINUE PPV AND VOLUME RESUS
FOLLOW TRAUMATIC CARDIAC ARREST
POST-ROSC MANAGEMENT (pg 7)

1. MAKE AN INCISION FROM THE MIDDLE OF THE STERNUM FOLLOWING THE FOURTH INTERCOSTAL SPACE TO THE THORACOSTOMY SITE EXTENDING TO POSTERIOR AXILLA LINE. DO NOT INCISE WITH SCALPEL PAST THE CHEST WALL MUSCLE TISSUE!
2. PLACE FINGER IN THORACOSTOMY (NAIL FACING LUNG TISSUE) THEN CUT INTERCOSTAL MUSCLE TOWARDS STERNUM FOLLOWING YOUR INCISION.
3. CUT THROUGH STERNUM WITH TUFFCUTS. HAVE GIGLI SAW SET AVAILABLE AND SET UP. USE SPENCER WELLS TO GUIDE GIGLI SAW WIRE BEHIND STERNUM IF REQ'D.
4. OPEN THE CHEST USING GAUZE OVER THE RIBS AND EXPOSE ORGANS FOR BELOW INJURIES. HAVE SUCTION AVAILABLE TO CLEAR THE FIELD IF REQ'D.

**CARDIAC
TAMPONADE**

EXPOSE HEART TO EXCLUDE or RELIEVE CARDIAC TAMPONADE BY CUTTING THE ANTERIOR PERICARDIAL SAC WITH AN INVERTED 'T' (AVOIDING THE PHRENIC NERVE) AND REMOVE HAEMATOMA BY HAND. PERFORM CUT TO INSPECT MYOCARDIUM FOR WOUNDS.

**CARDIAC
LACERATION**

INSPECT MYOCARDIUM & MANAGE CARDIAC WOUNDS BY USING SKIN STAPLES AS FIRST LINE, THEN CONSIDER FOLEY CATHETER.

**LUNG
HAEMORRHAGE**

CLAMP HILUM WITH FOLEY CATHETER AS A LIGATURE OR ROTATE LUNG ON ITS PEDICLE (LUNG TWIST) TO OCCLUDE HAEMORRHAGE

VF RHYTHM

FLICK MYOCARDIUM OR CLOSE CHEST AND SHOCK EXTERNALLY

**AORTIC
OCCLUSION**

LEFT HAND GOES UNDER LEFT LUNG UNTIL VERTEBRAE IS FELT. COMPRESS ANTERIOR SOFT TISSUE TO OCCLUDE DESCENDING AORTA.

**PEA OR
ASYSTOLE**

NO MYOCARDIAL ACTIVITY NEEDS CARDIAC MASSAGE & CONTINUE BLOOD PRODUCT RESUSCITATION UNTIL GOOD CARDIAC ACTIVITY IS SEEN

IF ROSC IS ACHIEVED, PATIENT WILL REQUIRE PHEA & RESUS UNTIL HANDOVER TO R2

CONSIDERATIONS FOR THORACOTOMY IN BLUNT TRAUMA

ULTRASOUND CONFIRMS PRESENCE OF CARDIAC TAMPONADE CAUSING CIRCULATORY ARREST STATE AND PATIENT IS SALVAGEABLE. CAN OCCUR IN UP TO 15% OF BLUNT TCA.

PATIENT DETERIORATES OR DOES NOT IMPROVE DESPITE HAEMORRHAGE CONTROL, BLOOD TRANSFUSION AND BILATERAL THORACOSTOMIES. USE FOR INTRATHORACIC OR PROXIMAL HAEMORRHAGE CONTROL. VERY LIKELY FUTILE IN PATIENTS WITH ASYSTOLE.